



THE SCOTS COLLEGE

2014

HSC Task 3 – In Class Test
10th June 2014
20%

Mathematics Extension 2

General Instructions

- Working time – 75 minutes
- Reading time – 5 minutes
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- Show all necessary working

Total marks – 43

Learning Intentions:

Polynomials

Integration

Volumes

QUESTION 1 - $\int (4 - x^2)^{-\frac{1}{2}} dx$

A) $\frac{(4 - x^2)^{\frac{1}{2}}}{-4x} + C$

C) $\ln \left| x + \sqrt{(4 - x^2)} \right| + C$

B) $-x(4 - x^2)^{\frac{1}{2}} + C$

D) $\sin^{-1} \frac{x}{2} + C$

QUESTION 2 - Which integral below gives the volume of the solid of revolution obtained by rotating the bounded region between $y = \sqrt{x}$, $x = 4$ and the line $y = 0$ around the y - axis ?

A) $\int_0^4 \pi x dx$

C) $\int_0^2 \pi(16 - y) dy$

B) $2 \int_0^4 \pi x \sqrt{x} dx$

D) $2 \int_0^2 \pi y dy$

QUESTION 3 - $\int x \sec^2 x dx$

A) $x \tan x + \ln |\cos x| + C$

C) $\frac{x^2}{2} \tan x + C$

B) $\frac{x^2}{2} (\sec^2 x - \tan^2 x) + C$

D) $\frac{\sec^3 x}{3x} + C$

QUESTION FOUR (12 MARKS) BEGIN A NEW SHEET OF PAPER

a) Evaluate

$$\int_0^{\frac{\pi}{6}} \cos \theta \sin^3 \theta \, d\theta \quad (2)$$

b) Find

$$\int \frac{\sqrt{x}}{1+x} \, dx \quad (3)$$

c) i) Express $\frac{x^2 + x + 2}{(x^2 + 1)(x + 1)}$ in the form $\frac{Ax + B}{x^2 + 1} + \frac{C}{x + 1}$ where A,B and C are constants. (2)

ii) Hence find $\int \frac{x^2 + x + 2}{(x^2 + 1)(x + 1)} \, dx$ (2)

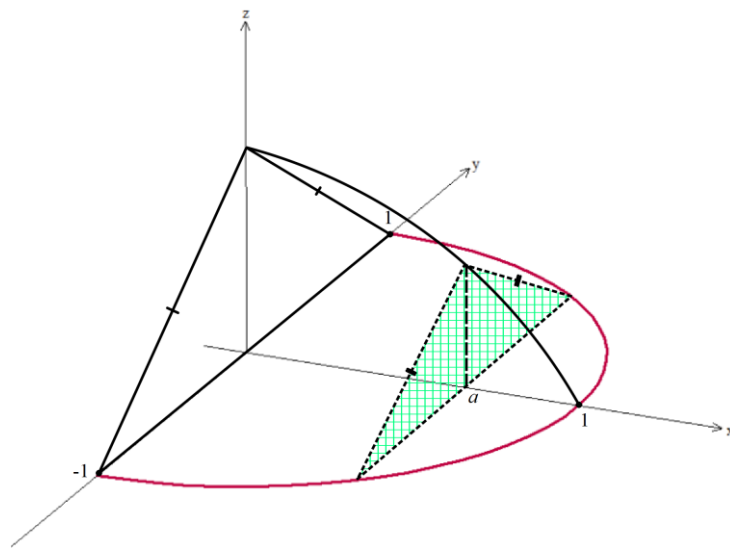
d) Using the substitution $t = \tan \frac{\theta}{2}$, evaluate, (3)

$$\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta + \cos \theta} \, d\theta$$

QUESTION FIVE (8 MARKS) BEGIN A NEW SHEET OF PAPER

- a) Given that the polynomial $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$ has a zero of multiplicity 3, find all the zeros of $P(x)$. (3)

- b) The base of a solid is the semi-circular region of radius 1 unit in the x - y plane as illustrated in the diagram below.



Each cross-section perpendicular to the x -axis is an isosceles triangle. Each of the two equal sides are three quarters the length of the third side.

- i) Show that the area of the triangular cross-section at $x = a$ is $\frac{\sqrt{5}}{2}(1-a^2)$ (3)
- ii) Hence find the volume of the solid. (2)

QUESTION SIX (10 MARKS) BEGIN A NEW SHEET OF PAPER

a) i) If $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$ for $n \geq 2$ prove that (3)

$$I_n = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1)I_{n-2}$$

ii) Hence evaluate (2)

$$\int_0^{\frac{\pi}{2}} x^2 \sin x \, dx$$

b) i) Prove that

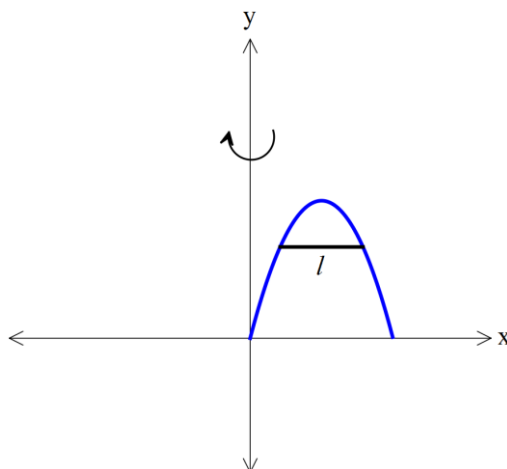
$$\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx \quad (2)$$

ii) Hence or otherwise, evaluate

$$\int_0^{\pi} x \sin^3 x \, dx \quad (3)$$

QUESTION SEVEN (10 MARKS) BEGIN A NEW SHEET OF PAPER

- a)** The region bounded by the curve $y = x - x^2$ and the x -axis is rotated around the y -axis to form a solid. When the region is rotated, the horizontal line segment l at height y sweeps out an annulus.

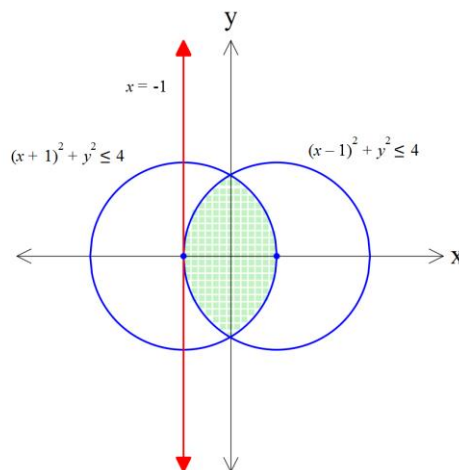


i) Show that the area of the annulus at height y is given by $2\pi\sqrt{\frac{1}{4} - y}$ (3)

ii) Find the volume of the solid (2)

- b)** The region enclosed by the circles $(x + 1)^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$ is rotated about the line $x = -1$.

Using the method of cylindrical shells,



i) Show that the volume is given by (3)

$$8\pi \int_0^1 \sqrt{4 - (x + 1)^2} dx$$

ii) Hence calculate the volume (2)

END OF EXAM

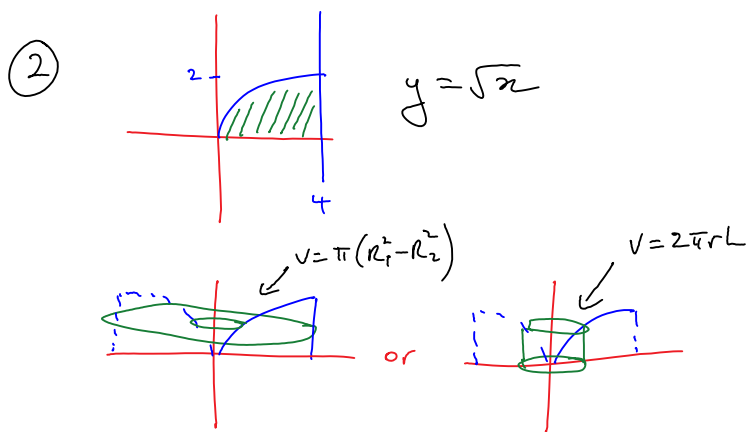
Ext 2 - task 3 Solutions - 2014

Thursday, 29 May 2014
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Multiple Choice

$$\begin{aligned} \textcircled{1} \quad \int (4-x^2)^{-\frac{1}{2}} dx &= \int \frac{1}{\sqrt{4-x^2}} dx \\ &= \sin^{-1}\left(\frac{x}{2}\right) + C \end{aligned}$$

D



$$\begin{aligned} \pi \int_0^2 (16-x^2) dy & \quad 2 \int_0^4 \pi x y \, dx \\ \pi \int_0^2 (16-y^4) dy & \quad 2 \int_0^4 \pi x \sqrt{x} \, dx \end{aligned}$$

$2 \int_0^4 \pi x \sqrt{x} \, dx$
✓

B

$$\begin{aligned} \textcircled{3} \quad \int x \sec^2 x \, dx &= x \tan x - \int \tan x \, dx \\ &= x \tan x - \int \frac{\sin x}{\cos x} \, dx \end{aligned}$$

← reverse chain

$$= x \tan x - \ln |\cos x| + C$$

A

Long Response

Q4.

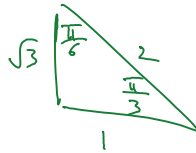
a) $\int_0^{\pi/6} \cos \theta \sin^3 \theta \, d\theta$

$$\left[\frac{1}{4} \sin^4 \theta \right]_0^{\pi/6}$$

$$\left[\frac{1}{4} \left(\sin \frac{\pi}{6} \right)^4 - 0 \right]$$

$$\left[\frac{1}{4} \left(\frac{1}{2} \right)^4 \right]$$

$$\boxed{\frac{1}{64}}$$



b) $\int \frac{\sqrt{x}}{1+x} \, dx$

let $u = \sqrt{x}$
 $u^2 = x$

$$\int \frac{u}{1+u^2} 2u \, du$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u \, du$$

$$2 \int \frac{u^2}{1+u^2} du$$

$$2 \int \frac{u^2+1-1}{u^2+1} du$$

$$2 \int 1 - \frac{1}{u^2+1} du$$

$$2 \left[u - \tan^{-1} u \right] + C$$

$$2\sqrt{x} - 2\tan^{-1}(\sqrt{x}) + C$$

c) i) $\frac{x^2+x+2}{(x^2+1)(x+1)} = \frac{Ax+B}{x^2+1} + \frac{C}{x+1}$

$$x^2+x+2 = (Ax+B)(x+1) + C(x^2+1)$$

∴

$$A+C = 1$$

$$A = 1-C \quad \dots \textcircled{1}$$

$$B+C = 2 \quad \dots \textcircled{2}$$

$$\text{let } x = -1$$

∴

$$2 = C + C$$

$$C = 1$$

$$A = 0$$

$$B = 1$$

∴

$$\frac{x^2+x+2}{(x^2+1)(x+1)} = \frac{1}{x^2+1} + \frac{1}{x+1}$$

ii)

$$\int \frac{1}{x^2+1} + \frac{1}{x+1} dx$$

$$\tan^{-1}(u) + \ln|x+1| + C$$

$$d) \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta + \cos \theta} d\theta$$

$$\tan \frac{\theta}{2} = t$$

$$\sin \theta = \frac{2t}{1+t^2}$$

$$\theta = 2 \tan^{-1} t$$

$$\cos \theta = \frac{1-t^2}{1+t^2}$$

$$\frac{d\theta}{dt} = \frac{2}{t^2+1}$$

$$\int_0^1 \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{t^2+1} dt$$

$$\begin{aligned} \theta &= \frac{\pi}{2} \\ t &= \tan \frac{\pi}{4} = 1 \\ \theta &= 0 \\ t &= 0 \end{aligned}$$

$$\int_0^1 \frac{2}{1+t^2+2t+1-t^2} dt$$

$$\int_0^1 \frac{2}{2t+2} dt = \int_0^1 \frac{1}{t+1} dt$$

$$= [\ln|t+1|]_0^1$$

$$= \ln 2 - \ln 1$$

$$= \ln 2$$

* Question 5:

$$a) P(x) = x^4 + x^3 - 3x^2 - 5x - 2$$

$$P'(x) = 4x^3 + 3x^2 - 6x - 5$$

$$P''(x) = 12x^2 + 6x - 6$$

$$= 6(2x^2 + x - 1)$$

$$= 6(2x - 1)(x + 1)$$

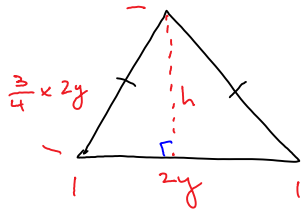
$$x = \frac{1}{2} \text{ or } -1$$

$P(x)$ is monic $\therefore (x+1)$ is triple root

$$P(x) = (x+1)^3(x-2)$$

b)

i)



$$x^2 + y^2 = 1$$

$$y = \pm \sqrt{1 - x^2}$$

$$h^2 = \left(\frac{3}{2}y\right)^2 - y^2$$

$$= \frac{9}{4}y^2 - \frac{4}{4}y^2 = \frac{5}{4}y^2$$

$$A = \frac{1}{2}bh$$

$$= \frac{1}{2} \times 2y \times \sqrt{\frac{5}{4}y^2}$$

$$A = y \times \frac{\sqrt{5}}{2} \times y$$

$$= \frac{\sqrt{5}}{2} (1 - x^2) \quad \text{at } x=a$$

$$A = \frac{\sqrt{5}}{2} (1 - a^2)$$

ii)

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 A \delta x$$

$$= \int_0^1 \frac{\sqrt{5}}{2} (1 - x^2) dx$$

$$= \frac{\sqrt{5}}{2} \left[x - \frac{x^3}{3} \right]_0^1$$

$$= \frac{\sqrt{5}}{2} \left[\left(1 - \frac{1}{3}\right) - 0 \right]$$

$$= \frac{\sqrt{5}}{3} \text{ units}^3$$

* Question 6

$$a) \quad I_n = \int_0^{\frac{\pi}{2}} \underbrace{x^n}_{\text{2nd}} \underbrace{\sin x}_{\text{1st}} dx$$

$$I_n = \left[-x^n \cos x \right]_0^{\frac{\pi}{2}} - n \int_0^{\frac{\pi}{2}} -x^{n-1} \cos x dx$$

$$I_n = 0 + n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x dx$$

$$I_n = +n \left(\left[x^{n-1} \sin x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} -(n-1)x^{n-2} \sin x dx \right)$$

$$I_n = +n \left(\left[+\left(\frac{\pi}{2}\right)^{n-1} \right] - (n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \sin x dx \right)$$

$$I_n = n\left(\frac{\pi}{2}\right)^{n-1} - n(n-1) I_{n-2}$$

$$ii) \quad \int_0^{\frac{\pi}{2}} x^2 \sin x dx = 2\left(\frac{\pi}{2}\right)^1 - 2(2-1) \int_0^{\frac{\pi}{2}} x^0 \sin x dx$$

$$\boxed{n=2}$$

$$= \pi - 2 \int_0^{\frac{\pi}{2}} \sin x dx$$

$$= \pi - 2 \left[-\cos x \right]_0^{\frac{\pi}{2}}$$

$$= \pi - 2 \left[0 - -1 \right]$$

$$= \pi - 2$$

b)

i)

$$\int_0^a f(x) dx$$

substitute

$$\text{let } x = a - u \Leftrightarrow u = a - x$$

$$dx = -du$$

$$\text{if } x = a \\ u = 0$$

$$\text{or} \\ x = 0 \\ u = a$$

$$\int_a^0 f(a-u) - du$$

$$- \int_a^0 f(a-u) du$$

$$\int_0^a f(a-u) du$$

substitute $u = x$

$$\int_0^a f(a-x) dx$$

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

ii)

$$\int_0^\pi x \sin^3 x dx$$

using part (i)

$$\int_0^\pi x \sin^3 x dx = \int_0^\pi (\pi - x) \sin^3(\pi - x) dx$$

$$= \int_0^\pi \pi \sin^3(\pi - x) dx - \int_0^\pi x \sin^3(\pi - x) dx$$

$$\sin(\pi - x) = \sin x$$

$$= \pi \int_0^\pi \sin^3(x) dx - \int_0^\pi x \sin^3(x) dx$$

$$2 \int_0^\pi x \sin^3 x dx = \pi \int_0^\pi \sin^3(x) dx$$

$$\int_0^\pi x \sin^3 x dx = \frac{\pi}{2} \int_0^\pi \sin^3 x dx$$

$$= \frac{\pi}{2} \int_0^\pi \sin^2 x \cdot \sin x dx$$

$$\text{let } u = \cos x$$

$$du = -\sin x dx$$

$$\begin{aligned}
 &= \frac{\pi}{2} \int_0^{\pi} \sin^2 x \cdot \sin x \, dx & du &= -\sin x \, dx \\
 &= \frac{\pi}{2} \int_1^{-1} (1-u^2) \cdot -du \\
 &= \frac{\pi}{2} \int_{-1}^1 (1-u^2) \, du \\
 &= \frac{\pi}{2} \left[u - \frac{u^3}{3} \right]_{-1}^1 = \frac{\pi}{2} \left[\frac{2}{3} + \frac{2}{3} \right] \\
 &= \frac{2\pi}{3}
 \end{aligned}$$

* Question 7

$$\begin{aligned}
 y &= \frac{1}{2} - \left(\frac{1}{2}\right) \\
 &= \frac{1}{2} - \frac{1}{4} = \frac{1}{4}
 \end{aligned}$$

a) $y = x - x^2$
 $= x(1-x)$

$$V = \int_0^{\frac{1}{4}} \pi (R_1^2 - R_2^2) \, dy$$

$$= \int_0^{\frac{1}{4}} \pi (x_1^2 - x_2^2) \, dy$$

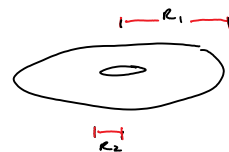
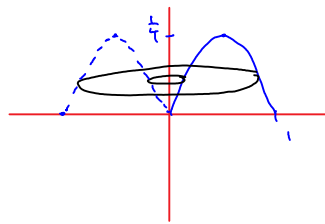
$$= \pi \int_0^{\frac{1}{4}} \left(\left(\frac{1}{2} + \sqrt{\frac{1}{4} - y} \right)^2 - \left(\frac{1}{2} - \sqrt{\frac{1}{4} - y} \right)^2 \right) dy$$

$$= \pi \int_0^{\frac{1}{4}} \left(\left(\frac{1}{4} + \sqrt{\frac{1}{4} - y} + \frac{1}{4} - y \right) - \left(\frac{1}{4} - \sqrt{\frac{1}{4} - y} + \frac{1}{4} - y \right) \right) dy$$

$$= \pi \int_0^{\frac{1}{4}} 2\sqrt{\frac{1}{4} - y} \, dy$$

$$= 2\pi \int_0^{\frac{1}{4}} \left(\frac{1}{4} - y \right)^{\frac{1}{2}} dy$$

$$= 2\pi \left[-\frac{2}{3} \left(\frac{1}{4} - y \right)^{\frac{3}{2}} \right]_0^{\frac{1}{4}}$$

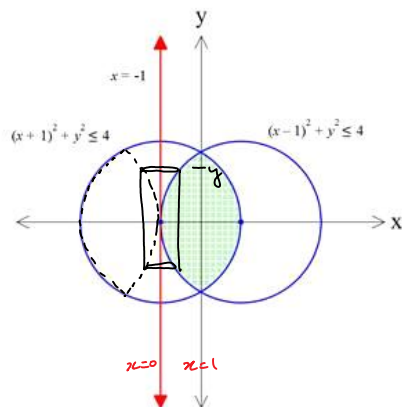


$$\begin{aligned}
 R_1 = x_1 &= \frac{1}{2} + \sqrt{\frac{1}{4} - y} \\
 R_2 = x_2 &= \frac{1}{2} - \sqrt{\frac{1}{4} - y}
 \end{aligned}$$

$$\begin{aligned}
 -y &= x^2 - x \\
 -y &= \left(x - \frac{1}{2}\right)^2 - \frac{1}{4} \\
 \frac{1}{2} \pm \sqrt{\frac{1}{4} - y} &= x
 \end{aligned}$$

$$\begin{aligned}
 &= 2\pi \left[-\frac{2}{3} \left(\frac{1}{4} - y \right)^{3/2} \right]_0^{1/4} \\
 &= 2\pi \left[0 - \left(-\frac{2}{3} \left(\frac{1}{4} \right)^{3/2} \right) \right] \\
 &= 2\pi \left[\frac{2}{3} \left(\frac{1}{8} \right) \right] \\
 &= \frac{\pi}{6} \text{ units}^3
 \end{aligned}$$

b)



$$\delta V = 2\pi r h \delta x$$

2 parts

$$V = \int_0^1 A_1 dx + \int_{-1}^0 A_2 dx$$

$$V = \int_0^1 2\pi(1+x) \sqrt{4-(x+1)^2} dx + \int_{-1}^0 2\pi(1+x) \sqrt{4-(x-1)^2} dx$$

$$V = 4\pi \int_0^1 (1+x) \sqrt{4-(x+1)^2} dx + 4\pi \int_{-1}^0 (-u+1) \sqrt{4-(-u-1)^2} du$$

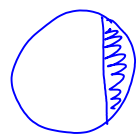
$(-u-1)^2 = (u+1)^2$

$$V = 4\pi \int_0^1 (1+x) \sqrt{4-(x+1)^2} dx + 4\pi \int_1^0 (u-1) \sqrt{4-(u+1)^2} du$$

$\int_a^b f(x) = -\int_b^a f(x)$

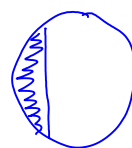
$$V = 4\pi \int_0^1 (1+x) \sqrt{4-(x+1)^2} dx + 4\pi \int_0^1 (1-u) \sqrt{4-(u+1)^2} du$$

$$(x+1)^2 + y^2 = 4$$



$$\int_0^1 A_1 dx$$

$$(x-1)^2 + y^2 = 4$$



$$\int_{-1}^0 A_2 dx$$

Remember - Radius & Height must be positive values.

Radius will always be bigger than 1.

$$R = 1 + x$$

Radius will always be smaller than 1 BUT x-value is negative
 $\therefore R = 1 + x$

Height for both volumes will be

$$H = 2y$$

$$V = 4\pi \int_0^1 (1+x) \sqrt{4-(x+1)^2} dx + 4\pi \int_0^1 (1-x) \sqrt{4-(x+1)^2} dx$$

$\int_a^b f(x) dx = - \int_b^a f(x) dx$

$$V = 4\pi \int_0^1 (1+x) \sqrt{4-(x+1)^2} dx + 4\pi \int_0^1 (1-x) \sqrt{4-(x+1)^2} dx$$

$x = u$

$$V = 4\pi \int_0^1 2 \sqrt{4-(x+1)^2} dx$$

Expand & simplify

$$V = 8\pi \int_0^1 \sqrt{4-(x+1)^2} dx \quad \text{as required}$$

ii)

$$V = 8\pi \int_0^1 \sqrt{4-(x+1)^2} dx$$

let $x+1 = 2\sin\theta$ when $x=1$ $\theta = \frac{\pi}{2}$
 $dx = 2\cos\theta d\theta$ $x=0$ $\theta = \frac{\pi}{6}$

$$V = 8\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sqrt{4-(2\sin\theta)^2} \cdot 2\cos\theta d\theta$$

$$= 8\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 4\cos^2\theta d\theta$$

$$= 32\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos 2\theta) d\theta$$

$$= 16\pi \left[\theta + \frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= 16\pi \left(\frac{\pi}{2} + 0 \right) - 16\pi \left(\frac{\pi}{6} + \frac{1}{2} \left(\frac{\sqrt{3}}{2} \right) \right)$$

$$= 16\pi \left[\frac{\pi}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{4} \right]$$

$$= 16\pi \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right]$$

$$= 11\pi^2$$

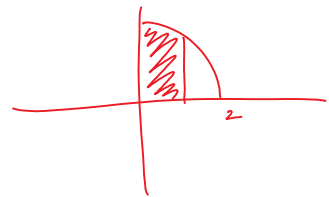
$$= \frac{16\pi^2}{3} - 4\pi\sqrt{3} \quad \text{units}^3$$

##

$$\text{ii)} \quad V = 8\pi \int_0^1 \sqrt{4 - (x-1)^2} \, dx$$

$$\text{let } x-1 = u$$

$$V = 8\pi \int_{-1}^0 \sqrt{4 - u^2} \, du$$



$$\text{let } u = 2\cos\theta$$

$$u=0 \quad \theta = \frac{\pi}{2}$$

$$du = -2\sin\theta \, d\theta$$

$$u=-1 \quad \theta = \frac{2\pi}{3}$$

$$= 8\pi \int_{\frac{2\pi}{3}}^{\frac{\pi}{2}} \sqrt{4 - 4\cos^2\theta} \cdot -2\sin\theta \, d\theta$$

$$= 8\pi \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} 2\sin\theta \cdot -2\sin\theta \, d\theta$$

$$= +32\pi \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \sin^2\theta \, d\theta$$

$$= +16\pi \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} 1 - \cos 2\theta \, d\theta$$

$$= +16\pi \left[\theta - \frac{1}{2}\sin 2\theta \right]_{\frac{\pi}{2}}^{\frac{2\pi}{3}}$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$= 1 - \sin^2\theta - \sin^2\theta$$

$$= 1 - 2\sin^2\theta$$

$$\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$= +16\pi \left[\theta - \frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{2}}$$

$$= 16\pi \left[\left[\frac{2\pi}{3} - \frac{1}{2} \left(\sin \frac{4\pi}{3} \right) \right] - \left[\frac{\pi}{2} - \frac{1}{2} \sin \pi \right] \right]$$

$$= 16\pi \left[\left(\frac{2\pi}{3} - \frac{1}{2} \left(-\frac{\sqrt{3}}{2} \right) \right) - \left(\frac{\pi}{2} - \frac{1}{2} (0) \right) \right]$$

$$= 16\pi \left[\frac{2\pi}{3} + \frac{\sqrt{3}}{4} - \frac{3\pi}{6} \right]$$

$$= 16\pi \left[\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right]$$

$$= \frac{8\pi^2}{3} + 4\sqrt{3}\pi$$